

Test task for analyst V3

Objective

Your role is to demonstrate advanced skills in database management, data extract, data manipulation, and visualization. You will handle multi-dimensional data, design suitable database schema, pull data from an external source and present data interactively using PowerBI.

Part 1: Data pull from external source

1. On the first step you should register here (https://apilayer.com/marketplace/exchangerates_data-api?#pricing) and generate free API token
2. Using Python write a script, that will collect currency exchange rates (for all currencies) and will save it to MySQL DB

Part 2: Database Management and Data Mocking

1. Design a MySQL database schema that includes the following tables:
 - a. exchange_rates (date, base_currency, target_currency, rate)
 - b. gdp (id, year, country, gdp_in_usd)
 - c. population (id, year, country, population)
2. Ensure to employ advanced SQL data types and constraints to maintain data integrity and reflect real-world scenarios (e.g., precision constraints in financial figures, etc.).
3. Use data that you got from API to exchange_rates table
4. Manually create entries for gdp and population tables. Data doesn't have to reflect real figures but must be coherent. Ensure diversity in the years and countries for a comprehensive analysis later.
5. Mimic the process of periodically fetching and updating exchange rates in your database by manually updating the exchange_rates table. It should contain rates for each country for a different years

Part 3: PowerBI Dashboarding

1. Establish a connection between your MySQL database and PowerBI.
2. Create a data model in PowerBI that merges the tables and recognizes the relationships between your datasets, particularly focusing on the keys that link GDP and population statistics to specific currency exchange rates.
3. Introduce calculated columns and measures using DAX (Data Analysis Expressions) to compute economic indicators, including:
 - a. GDP per capita
 - b. Normalization of GDP figures against various exchange rates (creating a dynamic conversion rate based on currency/year selection)
4. Design an interactive PowerBI dashboard that illustrates the GDP/GDP per capita by country. It must have the capability to display amounts recalculated in different currencies based on the available exchange rates in the database.
5. Implement advanced filtering features allowing users to refine the data displayed on the dashboard by various dimensions, including Country, Year, and Currency.
6. Ensure your PowerBI report elements follow best practices for data visualization. Elements should be logically arranged, aesthetically pleasing, and interactive, enhancing the end-user's experience and understanding of the data presented.

Part 4: Documentation and Justification

1. Send python code with comments inside that will explain the logic of your code
2. Accompany your database schema and PowerBI dashboard with a brief report that:
 - a. Explains the rationale behind your database design, including the choice of data types and constraints.
 - b. Describes any assumptions or financial/economic logic used in the computation of the indicators.
 - c. Suggests how the process could be automated and kept up-to-date in a real-world scenario (e.g., integrating live API data, database triggers, or scheduled ETL processes).

Part 5: Google Sheet & Data analysis

1. Data for work are available at the link
<https://docs.google.com/spreadsheets/d/1QZ5LGNvX0zJwOopT2bJI3TBWAXEBKEx0GLHmNN6ohOo/edit#gid=991196368>
2. Copy the Google Sheet table, and perform all manipulations in the copied file
3. Review the data, try to understand it yourself and normalize it
4. To normalize the data, it is not allowed to correct or split it manually. All actions must be done using formulas or Google Apps Script code
5. Your formulas and code in the final should be viewable
6. After normalizing the data, try to understand the insights based on the data. To do this, you can build graphs and charts, pivot tables etc.
7. Describe all graphs and charts, pivot tables etc. you created
8. Prepare detailed explanation of all insights you found
9. As result of this part send us back Document with expectations and final Google Sheet file

Notes: In the google Sheet table there is correlation between columns Browsing Path, Page Interaction, Time per Page. You should split data in these columns and connect data based on their position inside cell